

Counting Reads

- Download and open 05_ReadCounting.qmd run the chunks through package installation and loading
- Reminder: If packages do not install automatically, google the package and copy the different installation codes from Bioconductor—one of those should work (and AI is good at troubleshooting)
- If on your own computer and you haven't already, download the Integrated Genome Viewer (IGV):

<https://software.broadinstitute.org/software/igv/download>

- try Java included versions first, and then Java not included if that version doesn't open correctly

The Integrated Genomics Viewer (IGV)

- interactive tool for visualizing genomic data
- can be used to view read alignments (BAM or SAM files)
- BAM files need to first be sorted in genome position order and then indexed (easy to do in IGV)

How counting works...

- 1) provide mapped reads (usually in BAM format) or pseudo-alignment (in the case of Salmon), and
- 2) an annotation file that provides the genome coordinates for each gene or feature (e.g., alternatively spliced isoforms if that's your jam)
- 3) count the number of mapped reads per gene/feature

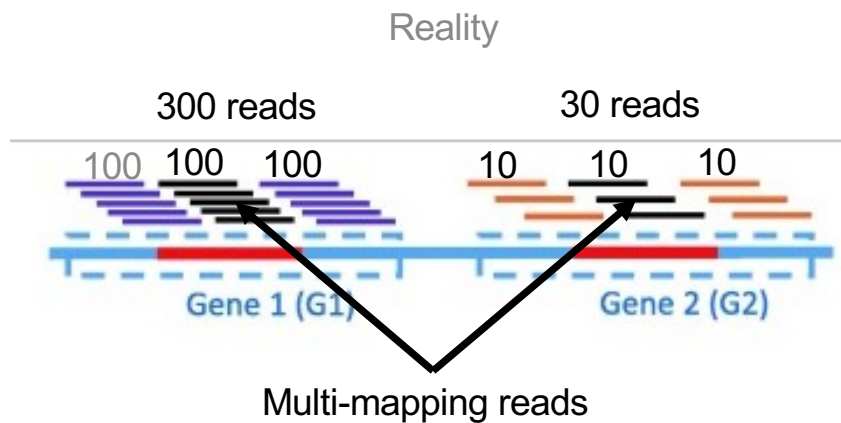
Annotation Files

- Files are General Feature Format (GFF) or General Transfer Format (GTF)
- Basically a key of feature names/attributes and genomic locations
- Can be downloaded from Ensembl, NCBI, organism databases

GTF Files

Column	Name	Example	Description
1	Chr.	I, II, III	Which chromosome
2	Source	ensembl	Database source
3	Feature	gene, exon, CDS	Type of feature
4	Start	335	Start position
5	End	649	End position
6	Score	.	Confidence score (often unused)
7	Strand	+, -	DNA strand
8	Frame	0, 1, 2	Reading frame for CDS
9	Attributes	gene_id "YAL069W"	Additional info

How to Handle Multi-Mapping Reads



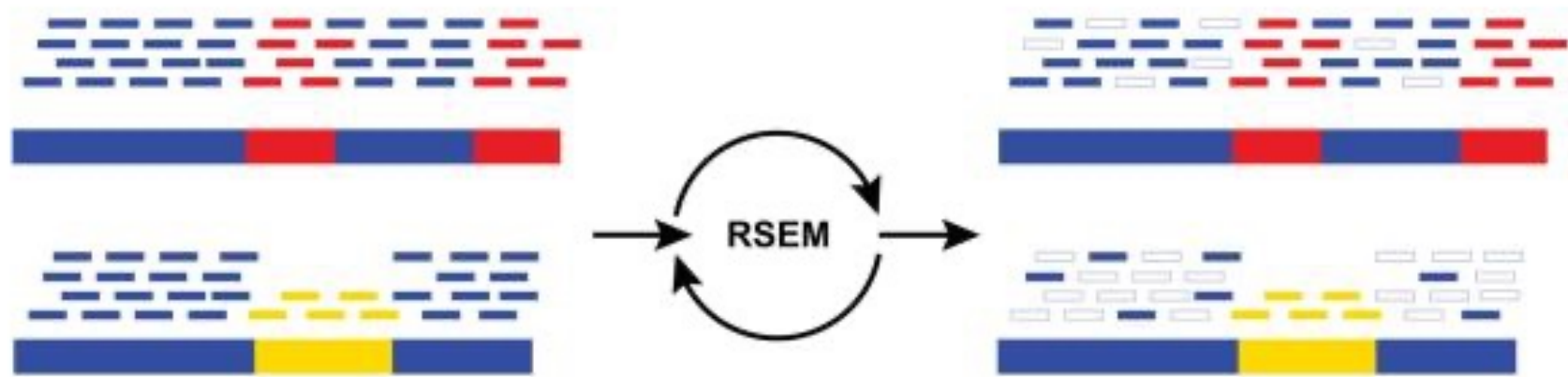
Approach to handle multireads	Read distribution representation	Counts
Ignore		G1: 200 G2: 20
Count once per alignment		G1: 310 G2: 130
Split them equally		G1: 255 G2: 75
Expectation-maximization		G1: 300 G2: 30

Adapted from: <https://www.sciencedirect.com/science/article/pii/S2001037020303032>

Expectation-Maximum (EM)

- Iteratively searches for maximum likelihood for parameter of interest in statistical models
- First performs an expectation (E) step, which generates likelihood estimate for a parameter (e.g., probability that a read comes from a transcript)
- Then performs a maximization (M) step, which evaluates whether changing those parameters improves the model.
- Then repeats the E and M steps until the model converges (i.e., stops changing)
- RSEM (RNA-Seq by Expectation Maximization) uses EM on mapped reads
- Salmon uses either EM or variational Bayesian EM (adds a probabilistic model to the maximum likelihood estimates)

Expectation-Maximum (EM)



- Blue = shared reads
- Red = unique to upper transcript
- yellow = unique to bottom transcript

Outputs from “Counting” Software

- Read counts for each feature (generally each gene, but can be isoforms if you care about alternative splicing)
- Normalized abundances within samples: Reads/Fragments per Kilobase Million (RPKM/FPKM) or Transcripts per Million (TPM)